MadLoop with TIRust

One-loop reduction · numerical agreement · runtime

CutTools · Ninja · COLLIER · NinjaRS · TIRust

Starting MadLoop revision `9228fc58e8f8` · one pinned CPU · full helicity/color sums

Main campaign acceptance: fail. The remaining campaign is incomplete. Every main process requires five-backend agreement, an upper 95% paired runtime-ratio bound no greater than 1.05, and passing existing-backend regression checks. Unmeasured cells are shown as —. A failed or incomplete row is not a speedup claim.

Runtime colour: green = faster, white = parity, red = slower. All displayed ratios are candidate / reference; smaller is faster.

Table 1: Main process matrix. Timings include the actual rescue work. Physics and runtime acceptance are separate.

Process · correction	Physics	Baseline µs/point	TIRust chain µs/point	Ratio · 95% interval	Existing-backend regressions
$u\bar{u} \to Z$ QCD · 2→1 · 100 points	pending	**63.2**	**71.1**	**×1.121** 1.103–1.139 fail	pending
$gg \to t\bar{t}$ QCD · 2→2 · 100 points	pending	—	—	— pending	pending
$u\bar{u} \to Zgg$ QCD · 2→3 · 100 points	pending	—	—	— pending	pending
$u\bar{u} \to ZZgg$ QCD · 2→4 · 100 points	pending	—	—	— pending	pending
$u\bar{d} \to W^+$ EW · 2→1 · 100 points	pending	—	—	— pending	pending
$e^+e^- \to \mu^+\mu^-$ EW · 2→2 · 100 points	pending	—	—	— pending	pending
$e^+e^- \to t\bar{t}\gamma$ EW · 2→3 · 100 points	pending	—	—	— pending	pending
$e^+e^- \to \mu^+\mu^-\gamma\gamma$ EW · 2→4 · 100 points	pending	—	—	— pending	pending

This study measures the new integration against the untouched starting branch, which already includes NinjaRS and earlier Fortran optimizations. It does not attribute those earlier improvements to TIRust.

1 Five-backend numerical agreement

The same saved momenta, model parameters, coupling orders and scale are evaluated by all five tools. The regression criterion is `3e-6`, with `1e-8` as the ordinary-point target. Born, finite, simple-pole and double-pole coefficients are retained. Near-zero poles use the existing MadLoop comparison policy. Final accepted DP/QP outcomes and failed evaluations are retained; the histograms do not enumerate internal rescue attempts.

Table 2: Comparison status and largest retained relative discrepancy. A dash or pending status denotes absent evidence.

Process	CutTools	Ninja	COLLIER	NinjaRS	TIRust
$u\bar{u} \to Z$ main · 100 points	**pending**	**pending**	**pending**	**pending**	**pending**
$gg \to t\bar{t}$ main · 100 points	**pending**	**pending**	**pending**	**pending**	**pending**
$u\bar{u} \to Zgg$ main · 100 points	**pending**	**pending**	**pending**	**pending**	**pending**
$u\bar{u} \to ZZgg$ main · 100 points	**pending**	**pending**	**pending**	**pending**	**pending**
$u\bar{d} \to W^+$ main · 100 points	**pending**	**pending**	**pending**	**pending**	**pending**
$e^+e^- \to \mu^+\mu^-$ main · 100 points	**pending**	**pending**	**pending**	**pending**	**pending**
$e^+e^- \to t\bar{t}\gamma$ main · 100 points	**pending**	**pending**	**pending**	**pending**	**pending**
$e^+e^- \to \mu^+\mu^-\gamma\gamma$ main · 100 points	**pending**	**pending**	**pending**	**pending**	**pending**
$u\bar{u} \to g\gamma$ control · 100 points	**pending**	**pending**	**pending**	**pending**	**pending**
$gg \to H$ control · 100 points	**pending**	**pending**	**pending**	**pending**	**pending**
$gg \to HH$ control · 100 points	**pending**	**pending**	**pending**	**pending**	**pending**

Agreement is assessed using independent numerical backends and higher precision when a discrepancy occurs. Known difficult scalar-box regions are retained as stress cases. A backend's own success flag does not establish cross-backend agreement.

2 Completed physical pilot checks

These eight separately retained checks each evaluate 100 saved physical points with all five reducers. They use the common strict stability policy and pass the `3e-6` regression criterion. Their timings are ineligible for performance acceptance, and they do not fill the pending timing-campaign rows above. An independent audit reparsed 4,800 evaluations and checked the retained inputs, executable provenance and raw statuses.

Table 3: Historical physical pilots, independent of the runtime gate. Residuals exclude poles accepted by the documented small-pole exception. All four existing reducers finally accepted 100 DP results in each row.

Process · 100 points	Five-backend regression	Ordinary target	Largest symmetric residual	TIRust accepted DP / QP
$u\bar{u} \to Z$	pass	pass	2.69×10^{-13}	100 / 0
$gg \to t\bar{t}$	pass	pass	4.58×10^{-12}	100 / 0
$u\bar{u} \to Zgg$	pass	miss	2.81×10^{-6}	87 / 13
$u\bar{d} \to W^+$	pass	pass	1.95×10^{-12}	100 / 0
$e^+e^- \to \mu^+\mu^-$	pass	pass	2.65×10^{-12}	100 / 0
$u\bar{u} \to g\gamma$	pass	pass	2.61×10^{-11}	100 / 0
$gg \to H$	pass	pass	1.55×10^{-10}	100 / 0
$gg \to HH$	pass	pass	4.61×10^{-9}	100 / 0

QCD 2→3 adjudication. The original comparison against CutTools DP misses the `1e-8` target at 28 points. CutTools, Ninja and TIRust QP agree within `9.89e-11` on four diagnostic points. Across all 28 target misses, the saved TIRust and COLLIER results meet the target against CutTools QP; CutTools DP misses all 28, while Ninja and NinjaRS miss only point 39. Every comparison still passes the regression criterion. These fixed-QP diagnostics are explicitly unchecked and do not establish automatic rescue stability.

The index `experiments/numerical/completed-pilots-index.json` binds 491 retained files and preserves the original comparisons. The detailed QP adjudication is in `experiments/qcd-n3-adjudication/`. Final accepted precision does not enumerate every internal reduction attempt. The remaining three large-process pilots are outside this frozen eight-case index.

3 Absolute timing by reduction backend

Values below are warmed full-amplitude times in µs per point. Initialization, compilation and model generation are excluded and recorded separately. All tools use matching pole, helicity/color and stability settings. Cache behaviour and actual rescue work remain part of the measurement. Recorded backend/precision histograms identify the final accepted reducer for every timed amplitude; they do not count every internal rescue attempt.

Table 4: Per-backend timing observations. Missing measurements are not represented as zero.

Process	CutTools	Ninja	COLLIER	NinjaRS	TIRust
$u\bar{u} \to Z$	—	—	—	—	—
$gg \to t\bar{t}$	—	—	—	—	—
$u\bar{u} \to Zgg$	—	—	—	—	—
$u\bar{u} \to ZZgg$	—	—	—	—	—
$u\bar{d} \to W^+$	—	—	—	—	—
$e^+e^- \to \mu^+\mu^-$	—	—	—	—	—
$e^+e^- \to t\bar{t}\gamma$	—	—	—	—	—
$e^+e^- \to \mu^+\mu^-\gamma\gamma$	—	—	—	—	—
$u\bar{u} \to g\gamma$	—	—	—	—	—
$gg \to H$	—	—	—	—	—
$gg \to HH$	—	—	—	—	—

4 Initialization and accepted precision

Each cell gives the first-amplitude initialization plus complete-stream warmup time in seconds, followed by the fraction of timed amplitudes finally accepted in QP. Process startup and parameter-card loading are outside the cold interval. A DP-only label refers to accepted amplitudes; it does not assert that no unsuccessful internal attempt occurred. The warmed timings on the previous page include all rescue work.

Table 5: Cold intervals and final accepted precision. Exact backend/precision counts and point-level results accompany the raw observations.

Process	CutTools	Ninja	COLLIER	NinjaRS	TIRust
$u\bar{u} \to Z$	—	—	—	—	—
$gg \to t\bar{t}$	—	—	—	—	—
$u\bar{u} \to Zgg$	—	—	—	—	—
$u\bar{u} \to ZZgg$	—	—	—	—	—
$u\bar{d} \to W^+$	—	—	—	—	—
$e^+e^- \to \mu^+\mu^-$	—	—	—	—	—
$e^+e^- \to t\bar{t}\gamma$	—	—	—	—	—
$e^+e^- \to \mu^+\mu^-\gamma\gamma$	—	—	—	—	—
$u\bar{u} \to g\gamma$	—	—	—	—	—
$gg \to H$	—	—	—	—	—
$gg \to HH$	—	—	—	—	—

5 Full versus poles-only tensors

10 of 12 predeclared settings completed seven alternating full/poles-only timing pairs; two DP settings were refused in both modes. All returned requested pole coefficients agree. These warmed tensor diagnostics use the historical release, with numerical caches disabled. They do not qualify full-amplitude performance.

Table 6: Median wall µs per call and geometric mean paired ratios. Green favors poles-only; red favors full evaluation. Intervals use seven paired log ratios and Student-t with six degrees of freedom.

Tensor · N / rank	Route	Prec.	Full µs	Poles µs	Poles / full · 95% interval
Massive tadpole N=1 · rank 6	Public Fortran	DP	9.01	9.42	**×1.041** 1.032–1.049
Massive tadpole N=1 · rank 6	Public Fortran	QP	1828	1838	**×1.005** 0.999–1.011
Massive bubble N=2 · rank 4	Generated adapter	DP	11.8	11.3	**×0.951** 0.939–0.964
Massive bubble N=2 · rank 4	Generated adapter	QP	1738	1121	**×0.646** 0.644–0.649
Infrared triangle N=3 · rank 3	Generated adapter	DP	8.95	8.78	**×0.977** 0.954–1.000
Infrared triangle N=3 · rank 3	Generated adapter	QP	1038	858	**×0.823** 0.815–0.832
Massive box N=4 · rank 4	Generated adapter	DP	32.2	18.0	**×0.558** 0.551–0.564
Massive box N=4 · rank 4	Generated adapter	QP	15384	2220	**×0.144** 0.141–0.146
Massive pentagon N=5 · rank 2	Generated adapter	DP	—	—	Both modes refused
Massive pentagon N=5 · rank 2	Generated adapter	QP	39078	2177	**×0.056** 0.055–0.057
Massive hexagon N=6 · rank 2	Generated adapter	DP	—	—	Both modes refused
Massive hexagon N=6 · rank 2	Generated adapter	QP	84852	5144	**×0.061** 0.060–0.061

Scope and observed limits. The rank-six tadpole calls the public Fortran bridge; the other tensors use the existing generated QCD 2⟶4 adapter, whose rank capacity is five. No generated rank limit was changed. The DP tadpole is about 4% slower in poles-only mode; QP is near parity. The massive N=5 and N=6 fixtures have zero poles, so their large QP gains do not describe infrared-singular pentagons or hexagons. DP refusals carry no timing claim.

Numerical caches are off; structural plans and scratch remain warm. Each measured batch lasted 0.825–1.925 s on pinned CPU 9. Driver tracing and coefficient output are outside the timer; input restoration, adapter mapping and bridge serialization remain inside. Shared-host compilation was active; host-load snapshots and CPU times are retained. These timings are explicitly ineligible for the full-amplitude acceptance gate.

Requested poles match within the declared DP/QP bounds; the largest scaled residual is `3.47e-18`. Omitted finite slots are NaN in the generated adapter and zero placeholders in the public bridge. The historical DSO is pinned by SHA256 `49af12f0b882…cd71ca12` and subsequent builds are outside this measurement. Raw batches, exact identities, and the independently audited 79-file inventory are retained under `experiments/poles-timing/`.

6 Methods and qualification

6.1 Physical samples and timing protocol

The main matrix contains QCD and electroweak corrections at every multiplicity from 2⟶1 through 2⟶4, using `loop_sm` and `loop_qcd_qed_sm_Gmu`. Each main case declares 100 deterministic points. On-shell 2⟶1 points vary their boost at the physical invariant; higher multiplicities use 1 TeV. Mixed QCD/QED and loop-induced cases supplement the main matrix. Threshold, soft/collinear, small-Gram and complex-mass cases are separate stress controls.

Performance acceptance uses at least seven alternating paired repetitions. The minimum batch target is one second; calibration aims for two seconds to accommodate shared-host speed variation. Actual durations are retained, and every batch must reach 95% of the minimum. The starting revision and candidate share their compiler, library installations, model/cards, point order and physical settings. Timing begins after the campaign's builds finish and uses one allowed CPU. The upper 95% paired ratio bound must be ≤1.05 separately for every main process. Averages cannot hide a failing row. A FULL–SKIP quantity, when reported in supporting data, is an **estimated reduction stage** rather than a pure tensor-kernel stopwatch.

All timing attempts are retained. Only an objectively short series with passing numerical checks can be retried within the same candidate campaign. A qualified runtime failure remains authoritative; a later favorable repetition cannot erase it. Source optimizations require a new candidate identity and campaign.

The materialized baseline chain is `8|6|7|1` the TIRust-first chain is `9|8|6|7|1`. Both run the same independent MadLoop stability policy.

The recorded cold interval includes first-amplitude initialization and warming the complete saved point stream; it excludes process startup and parameter-card loading. Numerical objects and linked archives are bound to the retained compiler recipes and hashes. The benchmark runs on a shared host; host-load observations accompany timing pairs, and this study does not claim exclusive use of the machine.

The common benchmark card sets `MLStabThres=1e-6` for both revisions and all reducers, with independent stability checks enabled. This policy was frozen before acceptance measurements after a native stress audit exposed correlated errors below the original threshold. Production retains its `1e-3` default. The tighter estimator does not replace the independent numerical truth gate or guarantee detection of every error.

6.2 Interface, poles and caches

MadLoop selects TIRust explicitly as `MLReductionLib=9`. A Fortran adapter transfers supplied invariants, complex squared masses and the squared renormalization scale to a separate Rust bridge. Tensor moments are permuted into MadLoop order and contracted in Fortran. The quadruple-precision path preserves binary128 inputs and uses MPFR working precision; double-double arithmetic is not labelled quadruple precision.

`LoopPoleMode` selects bare-loop UV/IR residues while preserving the ordinary finite coefficient. The distinct poles-only diagnostic path omits finite scalar roots and finite tensor arithmetic; it preserves double-pole contributions to simple poles and scaleless UV/IR separation. Partial results cannot be used as complete physical amplitudes. Cache and precision stores are separated by evaluation mode, and stability rechecks use independent values.

Qualification	Status	Evidence
CLI installation and ABI	**pass**	Final release installation and native DP/QP, pole/cache smoke pass. A later configuration-failure rollback fix passes 66 installer/control tests without changing native artifacts.
Generated Fortran integration	**pass**	17 final-release forced-mode runs cover external, relocated internal and NLO outputs; maximum symmetric error 1.29e-15. Forced modes do not establish stability acceptance.
QP transport and cache isolation	**pass**	Nine native bridge tests include binary128 last-bit transport, 256/384-bit convergence, failed-cache cleanup, bounded QP caching and rank-six mapping. Debug builds; no runtime claim.
Accelerated poles-only path	**pass**	N=1–6, rank six, UV/IR/scaleless and rescue/cache checks pass; finite evaluators and arithmetic are absent. 288 scalar comparisons and byte-identical regeneration pass.
Native library coexistence	**pass**	Both link orders and both initialization orders pass with unchanged NinjaRS dependency revisions. Earlier concurrent-license failure retained; qualification runs are serialized.

Qualification	Status	Evidence
Generated pole diagnostics	**partial**	96 served full/partial DP/QP pairs agree; 12 DP pairs refuse both modes and all 54 QP pairs pass. Partial finite outputs are NaN. Explicit refusals are retained and are not successful accelerated evaluations.
Generated cache lifecycle	**pass**	59 native records cover DP/QP references, hits, independent checks, modes, topology limits, byte budgets, clear/finalize and reinitialization. Stored and uncached tensors agree at the recorded precision.
Fortran prefix optimization	**pass**	481,536 bit/IEEE checks pass at each optimization level. Kernel timing improves; stubbed-wrapper gains vary. Physical-amplitude acceptance is separate.
Default stability stress case	**fail**	Default 1e-3 native checks miss one correlated rank-five DP error. At 1e-6 the production rescue block with actual tensor callbacks switches to QP and resolves it; complete physical initialization is outside this isolated check.
Complex-mass stress campaign	**partial**	All five reducers pass 19 moderate points; original extreme automatic results remain unqualified. Separate QP diagnostics identify legacy CutTools scalar-cutoff and DP-input limitations; private corrections agree with TIRust within 2.11e-11 and an independent double-pole formula.
Physical stress campaign	**fail**	24 primary requests give 12 checked completions, 11 rejected/nonfinite streams and one warmup timeout. QP resolves several discrepancies; the EW threshold comparison remains unresolved, and the soft-collinear Born result is nonfinite in every reducer.
Full five-backend acceptance	**pending**	The main matrix above records the independent physical and paired runtime gates. Completed interface checks do not substitute for those measurements.

6.3 Measured Fortran change

The selected optimization computes ordered momentum prefixes once, retaining the original zero seed and addition order. For six propagators it reduces prefix additions from 60 to 20. At each of `-O0` and `-O3`, 481,536 comparisons passed bitwise output and IEEE-exception checks across DP, QP and four rounding modes. Isolated six-propagator kernel median paired ratios were 0.443/0.347 (DP/QP, `-O0`) and 0.703/0.356 (`-O3`). These are shared-host microbenchmarks, not physical-amplitude speedups. Full-wrapper measurements were smaller and less consistent. Raw observations and source are retained in `experiments/fortran-prefix/`.

6.4 Retained numerical limitations

The bridge's 66 DP/QP stress evaluations include 14 explicit DP refusals and one served DP tensor coefficient above the diagnostic `3e-6` criterion. All 33 QP evaluations pass. For the failing massive rank-five box, native MadLoop reversal checks at the default `MLStabThres=1e-3` can accept correlated DP errors. At `1e-6`, the actual production rescue block with real tensor callbacks switches to QP and resolves the discrepancy below `1.2e-15`. The fixture, failing components, native check values and QP adjudication remain under `qualification/tirust-bridge/stability-audit/`. This isolated control-flow test omits physical helicity/filter initialization and does not replace matrix-element comparisons.

Pole-only tests prove that finite evaluators and finite tensor arithmetic are skipped. This does not imply every scalar topology is faster: illustrative warm scalar measurements include slower simple massless triangles and boxes, alongside faster massive and soft examples. A completely collapsed massless scalar box returns an explicit error. No unsupported request is counted as accelerated execution.

6.5 Reproduction and provenance

The accompanying `cases.json` fixes the matrix and thresholds; `summary.json` is derived from retained observations. The source plan, baseline manifest, dependency revisions, build environment, and raw measurement paths accompany this report. Rebuilding the PDF does not rerun numerical measurements.

Component	Recorded environment
Baseline	`9228fc58e8f8e48dadc70a6fcfb6617845aaeb1f`
Rust	rustc 1.97.1 (8bab26f4f 2026-07-14) (built from a source tarball)
Fortran	GNU Fortran (GCC) 15.2.0
Typst	typst 0.15.1 (9dfd3a08)
TIRust	`854b79dcb7fffcb35e92b4a8a3dd443369b798c4`
oneloopmaster	`bd703d09a78f7849a6b86a07096e6f222fc01193`

Presentation follows the pyAmpliCol report at revision 7266c1600f6e: its page geometry, fonts, navy headings, compact tables and logarithmic timing palette. All numerical measurements and conclusions in this report belong to the present MadLoop integration.